SOFTWARE PROJECT MANAGEMENT

STATEMENT OF WORK (SOW)

For

AuralAid: Adaptive AI Hearing with Contextual Understanding

Prepared By

Section Name: B

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1. Statement of Work

The Statement of Work (SOW) defines the scope, objectives, deliverables, timeline, and responsibilities for the **AuralAid** project. This document ensures alignment between stakeholders and provides a structured framework for planning, execution, and evaluation of the project.

1.1 Introduction

AuralAid is an AI-powered smart hearing aid system designed to improve speech clarity, provide real-time captions, detect emotions in speech, and generate safety alerts for high noise levels. Unlike traditional hearing aids that only amplify sound, AuralAid focuses on contextual understanding and accessibility using low-cost hardware and intelligent software techniques.

Table 1: Project Name

Field	Description
Project Name	AuralAid: Adaptive AI Hearing with Contextual Understanding

Table 2: Stakeholder Contact

Role	Name	Contact
Sponsor	HITEC University Taxila	(+92) 051 4908146
Supervisor	Mr. Junaid Ali Khan	HITEC University
Project Manager	Ayesha Nazir	22-SE-079
Team Member	Hamza Moeen	22-SE-025
Stakeholders	Hearing-impaired users, Supervisor, Evaluators	—

Background of Project
Hearing impairment affects millions worldwide, yet advanced hearing aids remain expensive and inaccessible in developing regions. Most existing solutions focus solely on amplification and perform poorly in noisy environments. Recent advancements in AI, DSP, and embedded systems enable intelligent hearing assistance at lower costs. AuralAid aims to leverage these technologies to provide an affordable, feature-rich solution.

Purpose of Project
The purpose of the AuralAid project is to design and develop a low-cost, AI-enabled hearing assistance system that enhances communication, safety, and situational awareness for hearing-impaired users through real-time captions, emotion detection, and noise alerts.

1.2 Scope of Work

Table 5: Scope

Item	To Be Done	Resources Needed	Outcome	Time Involved	General Steps
Item #1	Audio acquisition & preprocessing	ESP32-S3, MEMS Mic	Clean audio stream	1 month	Audio capture, filtering
Item #2	Noise reduction & DSP	ESP-DSP, C/C++	Improved speech clarity	1.5 months	Implement DSP filters
Item #3	Speech-to-text integration	AI models, Mobile App	Real-time captions	1.5 months	STT integration
Item #4	Emotion detection	ML models	Emotion indicators	1 month	Model training & testing
Item #5	Noise safety alerts	Sound sensors	User alerts	0.5 month	Threshold detection
Item #6	Mobile application	Flutter	Caption & alert UI	2 months	UI & BLE integration
Item #7	System integration	BLE, Firmware	End-to-end system	1 month	Integration & testing
Item #8	Testing & optimization	Test tools	Stable prototype	1 month	QA & optimization

1.3 Location

Table 6: Locations

Location	Site Duration
HITEC University Lab	8 months
Remote Development	8 months

1.4 Schedule

Table 7: Schedule

Phase	Start Date	Finish Date
Initiation & Planning	Sep 2024	Oct 2024
Requirements & Design	Oct 2024	Dec 2024
Hardware & DSP Development	Jan 2025	Feb 2025
AI & App Development	Feb 2025	Mar 2025
Integration & Testing	Mar 2025	Apr 2025
Finalization & Documentation	Apr 2025	May 2025

1.5 Deliverables

Table 8: Deliverables

Deliverable	Objective	Due Date
Proposal & SOW	Project approval	Oct 2024
SRS & Architecture	Requirements definition	Dec 2024
Hardware Prototype	Functional audio system	Feb 2025
AI & Mobile App	Captions & alerts	Mar 2025
Final Prototype	Fully integrated system	Apr 2025
Documentation	Thesis & manuals	May 2025

1.6 Milestones

Table 9: Milestones

Milestone	Start	End
Title Defense	Sep 2025	Oct 2025
SRS Approval	Oct 2025	Dec 2025
DSP Completion	Jan 2026	Feb 2026
AI Integration	Feb 2026	Mar 2026
Final Evaluation	Apr 2026	May 2026

1.7 Tasks

Table 10: Tasks

Task	Duration	Responsible
Feasibility Study	3 days	Ayesha Nazir
SOW & Proposal	4 days	Hamza Moeen
Requirement Analysis	7 days	Whole Team
DSP Development	30 days	Hamza Moeen
AI Model Integration	25 days	Ayesha Nazir
Mobile App Development	35 days	Whole Team
Testing & QA	30 days	Whole Team
Documentation	60 days	Ayesha Nazir

1.8 Standards and Testing

Table 12: Standards and Testing

Test Type	Responsibility	Due Date
Unit Testing	Team	Feb 2025
Integration Testing	Team	Mar 2025
Performance Testing	Team	Apr 2025
Usability Testing	Users	Apr 2025

1.9 Define Success

- Accurate real-time captions
- Improved speech clarity in noisy environments
- Reliable noise alerts
- Positive supervisor and user feedback
- On-time project completion

1.10 Requirements

Table 14: Requirements

Category	Details
Hardware	ESP32-S3, MEMS microphones, smartphones
Software	Arduino IDE, Flutter, TensorFlow Lite
Tools	GitHub, Figma, Jira

1.12 Payments

Table 16: Payments

Vendor	Cost	Due
Hardware Components	PKR 30,000	During development

1.13 Closure

Table 17: Closure

Document	Signed Off
Final Report	Supervisor
Prototype Approval	Evaluators
Documentation	Project Manager